

Supplemental Material for “The UZH protocol: Separating and reducing Gaussian-basis and pseudopotential errors in CP2K”

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I. SOURCE MATERIAL AND IMPLEMENTATION DETAILS

The main text summarizes the protocol, while this Supplemental Material records additional details that are useful for reproduction and for future development of the UZH settings. The organization follows the original GTH and MOLOPT literature [1–3], the revised MOLOPT benchmarking work [4], the CP2K implementation papers [5, 6], and the AiiDA verification protocol and its Supplemental Information [7, 8].

A. CP2K ATOM and basis optimization

The CP2K source tree contains a dedicated ATOM input section for atomic calculations. The relevant run types are energy calculations, basis optimization, and pseudopotential optimization. The ATOM method section supports restricted and unrestricted Kohn-Sham and Hartree-Fock atomic calculations and scalar-relativistic Hamiltonians including ZORA-type and Douglas-Kroll-Hess variants. The potential section supports all-electron, GTH, UPF, SGP, and ECP potentials. For the GTH format, the input layer exposes a CP2K-standard GTH potential block, and the print layer can write the optimized potential in CP2K and UPF-like forms.

The pseudopotential optimizer constructs both all-electron and pseudopotential atomic references. Occupations are partitioned into core and valence contributions; semicore choices are therefore explicit in the input. The optimizer evaluates the all-electron reference with the selected scalar-relativistic Hamiltonian, constructs the corresponding pseudopotential atom, and fits the compact GTH parameters with a derivative-free Powell algorithm. This machinery is what allows the UZH protocol to improve the pseudopotential component once the SIRIUS-GTH-UZH versus SIRIUS-FP-LAPW comparison identifies a pseudopotential-limited element.

The CP2K basis optimizer works from template, work, and output basis files. It supports simultaneous fitting of related basis levels and allows the condition number of the overlap matrix to enter the objective. This mirrors the original MOLOPT design: diffuse flexibility is desirable for weak interactions and response properties, but the final basis must remain numerically stable in condensed-phase simulations.

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B. SIRIUS calculations

SIRIUS is used in two complementary modes [9–11]. In pseudopotential plane-wave mode, SIRIUS-GTH-UZH uses the same GTH-UZH pseudopotentials as CP2K but removes the localized Gaussian-basis approximation. In all-electron FP-LAPW mode, SIRIUS provides the reference equation of state. The two SIRIUS modes therefore isolate the pseudopotential error, while the CP2K-GTH-UZH versus SIRIUS-GTH-UZH comparison isolates the CP2K Gaussian-basis error.

II. MOLECULAR CALIBRATION

The molecular calibration assesses the revised UZH-MOLOPT basis sequence before the solid-state verification step. The benchmark contains small molecules covering main-group bonds, polar bonds, non-linear and dihedral angles, dipole moments, polarizabilities, and vibrational frequencies. The role of this stage is to ensure chemically balanced transferability and numerical stability before the basis is used in periodic calculations.

III. REPRESENTATIVE EQUATION-OF-STATE CURVES

The periodic-table heat maps in the main text are compact summaries of many equation-of-state calculations. Representative raw equation-of-state comparisons are shown here for elements that illustrate different error regimes.

IV. RECOMMENDED ORGANIZATION OF RELEASED DATA

For publication, the repository accompanying the paper should contain:

1. CP2K-GTH-UZH AiiDA workflow inputs and parsed results.
2. SIRIUS-GTH-UZH AiiDA workflow inputs and parsed results.
3. SIRIUS-FP-LAPW all-electron workflow inputs and parsed results.
4. Optimized UZH-MOLOPT basis files and GTH-UZH pseudopotential files.
5. Scripts used to evaluate the ε metric and generate the periodic-table maps.

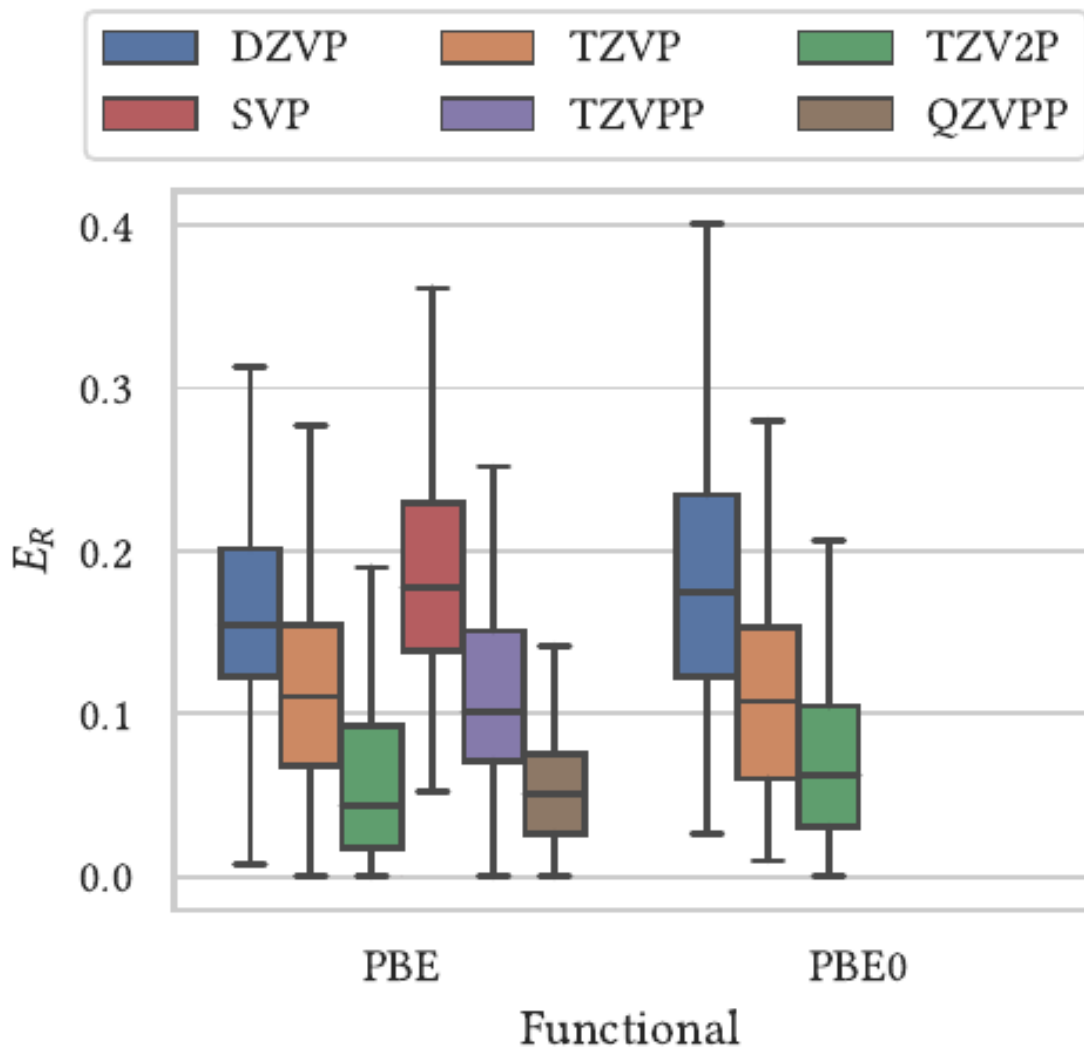


FIG. 1. Additional molecular benchmark statistics for the UZH-MOLOPT basis sequence.

6. A manifest documenting code versions, pseudopotential hashes, basis-file hashes, and workflow protocol names.

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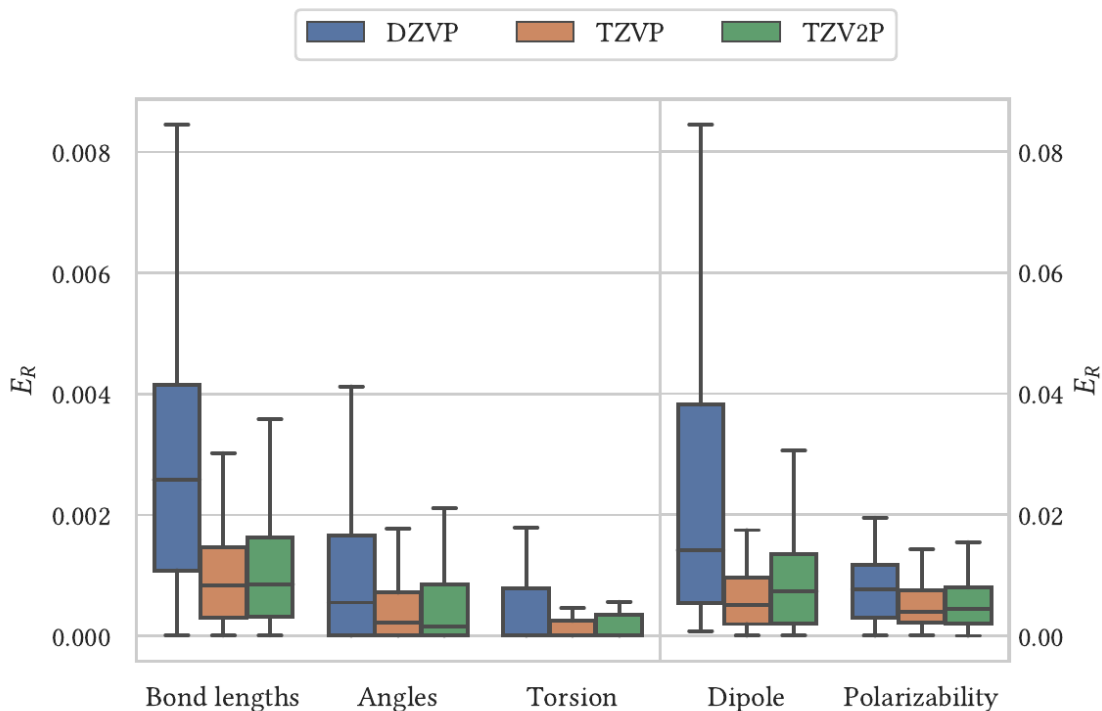


FIG. 2. Additional molecular benchmark statistics for properties that are more sensitive to basis flexibility.

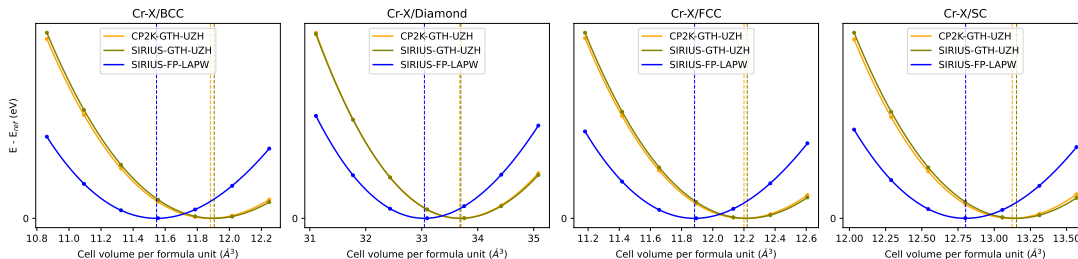


FIG. 3. Representative equation-of-state curves for Cr. The behavior is characteristic of a pseudopotential-limited case in the original settings.

systems in gas and condensed phases, *The Journal of Chemical Physics* **127**, 114105 (2007).

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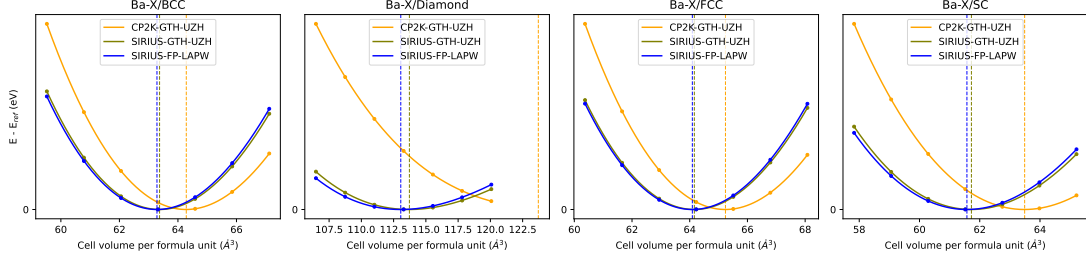


FIG. 4. Representative equation-of-state curves for Ba. The comparison highlights a case with a substantial Gaussian-basis contribution in the original CP2K-GTH-UZH settings.

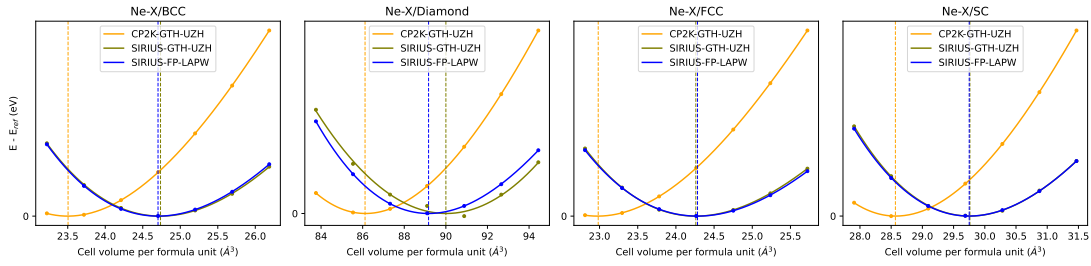


FIG. 5. Representative equation-of-state curves for Ne. Noble gases are sensitive tests of diffuse basis flexibility because their equation-of-state curves are shallow.

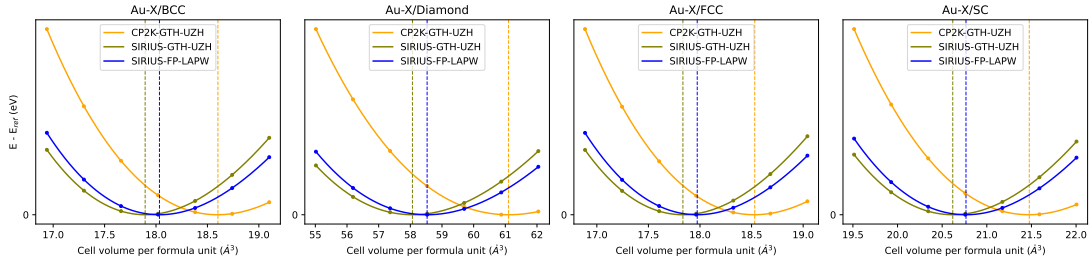


FIG. 6. Representative equation-of-state curves for Au, illustrating the behavior of a heavy element where relativistic pseudopotential transferability and basis completeness must both be monitored.

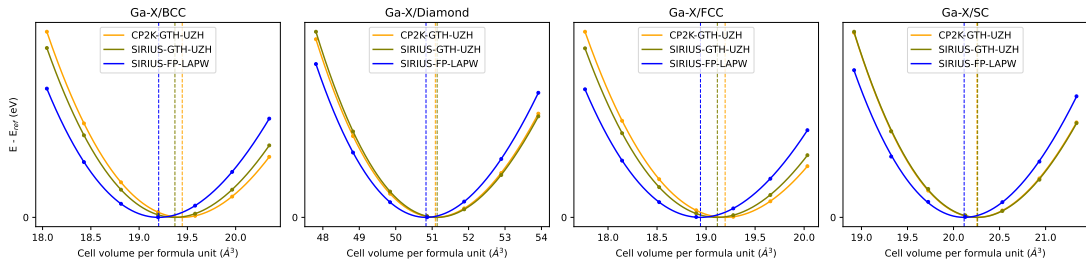


FIG. 7. Representative equation-of-state curves for Ga.

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